

Spectroscopy as a Probe of Spacetime Foam

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Quantum-gravitational spacetime foam imprints an achromatic, stochastic arrival-time jitter on light from distant sources. We show that this jitter modifies the first-order temporal coherence $g^{(1)}(\tau)$ and hence the observed spectrum, providing a spectroscopic probe complementary to existing imaging-channel (PSF, fringe-visibility, Strehl) constraints. In the narrow-band limit the spectrum factorizes into a coherent line core suppressed by $e^{-\sigma_\phi^2}$ and an incoherent pedestal, yielding a spectral-Strehl bound on the Ng–Perlman foam exponent α . Applied to the MeV emission lines of GRB 221009A we obtain $\alpha \gtrsim 0.64$, and to the redshifted Fe K α line of the Compton-thick AGN BzK 4892, $\alpha \gtrsim 0.58$. For the broadband CMB the foam induces an $e^{-\sigma_\phi^2(\nu)}$ deficit ($\sigma_\phi^2 \propto \nu^2$) with a low-frequency pedestal; a Fisher analysis of COBE/FIRAS reaches $\alpha \lesssim 0.57$, with PIXIE projected to extend coverage to $\alpha \lesssim 0.60$. None of these bounds exclude the holographic value $\alpha = 2/3$, which would require a confirmed narrow spectral line at $E_{\text{obs}} \gtrsim 720$ MeV in a $z \sim 0.15$ GRB.

I. INTRODUCTION

A long-standing expectation from the union of quantum mechanics and general relativity is that the smooth pseudo-Riemannian manifold of classical spacetime breaks down at the Planck length $\ell_P = \sqrt{\hbar G/c^3} \simeq 1.616 \times 10^{-35}$ m. In Wheeler’s original “spacetime foam” picture [1], quantum fluctuations of the metric on this scale render the geometry stochastic; similar conclusions are reached in the Euclidean path-integral treatment [2] and in modern holographic and entropic-gravity arguments from very different starting points. While a complete theory of quantum gravity remains elusive, the question of whether residual imprints of Planck-scale geometric fluctuations can accumulate over cosmological baselines and become observable is one of the few empirical handles on physics in the Planck regime, and it has motivated a substantial body of phenomenological work over the past two decades [3, 4].

A widely studied parametrization due to Ng and collaborators [4–6], the α -foam model, encodes the foam dynamics in a power-law path-length variance

$$\sigma_\ell^2(D) = A_\alpha D^{2(1-\alpha)} \ell_P^{2\alpha}, \quad (1)$$

where D is the proper distance to the source, α parametrizes the foam model, and A_α is a dimensionless coefficient of order unity. A photon traversing distance D passes through approximately $N \sim D/\ell_P$ independent Planck-scale spacetime cells, each imparting a tiny random displacement to the photon path; the exponent α controls how the per-cell displacements accumulate: for random-walk foam ($\alpha = 1/2$), contributions add in quadrature ($\sigma_\ell \propto \sqrt{D\ell_P}$); holographic foam ($\alpha = 2/3$) grows more slowly, constrained by an area-law bound on the number of independent degrees of freedom per Planck area; and conservative foam ($\alpha = 1$) yields a distance-independent $\sigma_\ell \propto \ell_P$, corresponding to perfectly correlated cell contributions.

The primary observational probe of stochastic foam to date has been the imaging channel. Spacetime foam accumulates a cumulative phase dispersion $\Delta\phi \simeq 2\pi\ell^{1-\alpha}\ell_P^\alpha/\lambda$ across the entrance aperture, degrading PSF sharpness and erasing interference fringes once $\Delta\phi \gtrsim 2\pi$ [7]. For PKS 1413+135 ($\ell = 1.216$ Gpc) observed with HST at $\lambda = 1.6 \mu\text{m}$, the random-walk model ($\alpha = 1/2$) predicts $\Delta\phi \sim 10 \times 2\pi$, so the detection of Airy rings in this image marginally rules out that model; the holographic model ($\alpha = 2/3$) predicts $\Delta\phi \sim 10^{-9} \times 2\pi$ and is obviously not ruled out [4, 8, 9]. Models with long-range transverse correlations evade even the random-walk bound [10].

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The most luminous GRB ever recorded, GRB 221009A at $z = 0.151$, has been examined in the imaging channel by Steinbring [11] and Steinbring & Ng [12]. Using a multiwavelength average of foam-induced angular broadening — analogous to a seeing disk from atmospheric turbulence — Steinbring [11] showed that the holographic ($\alpha = 2/3$) model predicts an angular widening below $\sim 1^\circ$, consistent with the Fermi and Swift localization data for GRBs at $z \leq 5$ up to peak energies of ~ 10 MeV. Steinbring & Ng [12] extended this to the full multiwavelength follow-up of GRB 221009A — from optical and near-infrared through X-rays and the highest-energy gamma-rays ever attributed to a GRB — computing the holographic-foam-widened PSF for real telescopes and finding all data consistent with $\alpha = 2/3$. These imaging-channel analyses show that the holographic model is not excluded by the spatial localization of GRB 221009A.

We develop a framework based on the first-order temporal coherence function $g^{(1)}(\tau)$ of light from astrophysical sources. This function is complementary to the first-order spatial coherence function $g^{(1)}(B)$, which forms the basis of imaging-channel probes. The first-order coherence — the field-field autocorrelation — inherits a foam suppression factor that decreases continuously from unity toward complete fringe erasure as the accumulated phase variance σ_ϕ^2 grows. In our framework the phase variance arises from temporal fluctuations of foam, whereas in the imaging channel it arises from spatial fluctuations; correspondingly, foam imprints itself on the spectrum rather than on the spatial brightness profile.

The remainder of the paper is organized as follows. Section II introduces the spacetime foam model: the arrival-time jitter parameterization, the temporal autocorrelation $C_\alpha(\tau)$, and the distribution of differential jitter. Section III derives the first-order coherence function $g_{\text{obs}}^{(1)}(\tau)$ of the observed field, establishes closed-form expressions for Lorentzian and Voigt sources, and identifies the narrow-band limit in which the foam suppression factor simplifies. Section IV translates these results into spectroscopic observables via the Wiener–Khinchin theorem and connects them to fringe visibility and the spectral Strehl ratio. Section V applies the spectral-Strehl bound to three sources: GRB 221009A (Section V A), the high-redshift galaxy BzK 4892 (Section V B), and the CMB as observed by COBE/FIRAS (Section V C). Section VI summarizes the main results and outlines future observational directions.

II. SPACETIME FOAM MODEL

A. Arrival-Time Jitter as a Time Reparameterization

We model the detected field as a stochastically time-warped version of the intrinsic source field:

$$E_{\text{obs}}(t) = E_{\text{src}}(t - \delta t(t)), \quad (2)$$

where $E_{\text{src}}(t)$ is a single spatial mode of a stationary zero-mean complex Gaussian process (chaotic light, appropriate for virtually all known astrophysical sources) and $\delta t(t)$ is the foam-induced arrival-time jitter, a stationary stochastic process statistically independent of E_{src} . The jitter arises from cumulative path-length fluctuations along the line of sight of propagation distance D :

$$\delta t(t) = \frac{1}{c} \Delta \ell_{\text{tot}}(t). \quad (3)$$

In the geometric-optics limit the path-length shift $\Delta \ell_{\text{tot}}$ is a purely geometric quantity and hence frequency-independent (achromatic), in contrast to plasma dispersion which scales as λ^2 . This achromaticity distinguishes foam from other astrophysical screens — such as plasma dispersion, interstellar scintillation, and atmospheric turbulence — all of which imprint frequency-dependent signatures. For broadband sources such as the CMB, the foam distortion is likewise distinct from standard spectral distortions (μ , y) that have different frequency dependences and are not generated by geometric path-length jitter.

B. Spatial and Temporal Fluctuations of Spacetime Foam

The scalar variance $\sigma_\ell^2(D)$ (Eq. (1)) characterizes the amplitude of foam-induced path-length fluctuations along a single ray; here we describe how those fluctuations are correlated across across time (temporal decorrelation).

The temporal evolution of spacetime foam — an aspect not specified by Eq. (1) — determines how the arrival-time jitter $\delta t(t)$ along a single line of sight is correlated across the lag τ . Quantum foam models posit that each Planck-scale cell along the line of sight fluctuates on a characteristic timescale τ_{foam} , so that the jitter at times t and $t + \tau$ is correlated for $\tau \ll \tau_{\text{foam}}$ and essentially independent for $\tau \gg \tau_{\text{foam}}$. We introduce $C_\alpha(\tau; D)$ as the normalized *temporal* autocorrelation of the per-cell path-length fluctuation, integrated over the $\sim D/\ell_P$ cells along the line of

sight. The timescale τ_{foam} is an additional free parameter of the foam model, not fixed by the α -model variance relation alone. If τ_{foam} is of order the Planck time, $\ell_P/c \approx 5 \times 10^{-44}$ s, then $C_\alpha(\tau) \approx 0$ for all accessible lags: the foam is fully decorrelated on any observable timescale, the most optimistic scenario for detection. The temporal covariance between two rays separated by in time τ takes the form

$$\langle \Delta \ell_{\text{tot}}(t) \Delta \ell_{\text{tot}}(t + \tau) \rangle = \sigma_\ell^2(D) C_\alpha(\tau; D). \quad (4)$$

C. Differential Jitter and Its Distribution

With the temporal covariance model in place, we define the key derived quantity that enters the coherence integrals. Define the differential jitter between a photons arriving times t and $t + \tau$:

$$\Delta(\tau) \equiv \delta t(t + \tau) - \delta t(t). \quad (5)$$

For fixed D , stationarity implies the statistics depend only on τ . Expanding the square and applying Eq. (??):

$$\begin{aligned} \sigma_\Delta^2(\tau) &= \frac{1}{c^2} \langle [\Delta \ell_{\text{tot}}(t + \tau) - \Delta \ell_{\text{tot}}(t)]^2 \rangle \\ &= \frac{1}{c^2} [\sigma_\ell^2 + \sigma_\ell^2 - 2\sigma_\ell^2 C_\alpha(\tau)] \\ &= \frac{2\sigma_\ell^2(D)}{c^2} [1 - C_\alpha(\tau; D)]. \end{aligned} \quad (6)$$

By the central limit theorem (CLT), $\Delta(\tau)$ is a sum of contributions from many foam cells and is therefore Gaussian with zero mean:

$$\Delta(\tau) \sim \mathcal{N}(0, \sigma_\Delta^2(\tau)), \quad (7)$$

where σ_Δ^2 is taken from Eq. (6). The Gaussian approximation follows from the CLT: for any source at $D \gtrsim 1$ kpc, the cell count $N_{\text{cells}} \sim D/\ell_P \gtrsim 10^{46}$ amply satisfies the large- N requirement and higher-cumulant corrections are negligible.

III. FIRST-ORDER COHERENCE

A. Jitter Convolution Formula for $g^{(1)}$

The observed first-order coherence is

$$g_{\text{obs}}^{(1)}(\tau) = \frac{\langle E_{\text{obs}}^*(t) E_{\text{obs}}(t + \tau) \rangle}{\langle |E_{\text{obs}}(t)|^2 \rangle}. \quad (8)$$

Substituting Eq. (2) and letting $u = t - \delta t(t)$, $v = t + \tau - \delta t(t + \tau)$, so that $v - u = \tau - \Delta(\tau)$, stationarity of E_{src} gives

$$\langle E_{\text{src}}^*(u) E_{\text{src}}(v) \rangle = \langle |E_{\text{src}}|^2 \rangle g_{\text{src}}^{(1)}(\tau - \Delta(\tau)). \quad (9)$$

Since δt is independent of E_{src} , averaging over the jitter yields

$$g_{\text{obs}}^{(1)}(\tau) = \langle g_{\text{src}}^{(1)}(\tau - \Delta(\tau)) \rangle_{\delta t} = \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma_\Delta^2(\tau)) g_{\text{src}}^{(1)}(\tau - \Delta) d\Delta. \quad (10)$$

The observed first-order coherence is the intrinsic coherence function convolved with the Gaussian differential-jitter kernel.

B. Closed Forms for Lorentzian and Voigt Sources

As a representative case admitting a closed-form result, we take a Lorentzian line profile, which corresponds to an exponential coherence function via the Wiener-Khinchin theorem:

$$g_{\text{src}}^{(1)}(\tau) = e^{-|\tau|/\tau_L}, \quad (11)$$

where τ_L is the coherence time. Evaluating Eq. (10) by completing the square and splitting on the sign of $\tau - \Delta$ gives, for $\tau > 0$:

$$g_{\text{obs}}^{(1)}(\tau) = e^{\sigma_\Delta^2/2\tau_L^2} \cdot \frac{1}{2} \left[e^{-\tau/\tau_L} \operatorname{erfc}\left(\frac{\sigma_\Delta/\tau_L - \tau/\sigma_\Delta}{\sqrt{2}}\right) + e^{\tau/\tau_L} \operatorname{erfc}\left(\frac{\sigma_\Delta/\tau_L + \tau/\sigma_\Delta}{\sqrt{2}}\right) \right]. \quad (12)$$

In the weak-jitter limit $\sigma_\Delta \ll \tau_L$, the Gaussian average of $e^{-|\tau-\Delta|/\tau_L}$ over $\Delta \sim \mathcal{N}(0, \sigma_\Delta^2)$ is dominated by $|\Delta| \ll |\tau|$, so one expands around $\Delta = 0$:

$$g_{\text{obs}}^{(1)}(\tau) \approx g_{\text{src}}^{(1)}(\tau) \left[1 + \frac{\sigma_\Delta^2(\tau)}{2\tau_L^2} + \mathcal{O}(\sigma_\Delta^4) \right]. \quad (13)$$

Here $\sigma_\Delta(\tau)$ is τ -dependent through $C_\alpha(\tau)$, and the expansion is in $\sigma_\Delta(\tau)/\tau_L$ at fixed τ . The correction is positive because $e^{-|\tau|/\tau_L}$ is convex and Jensen's inequality applies. The normalization $g_{\text{obs}}^{(1)}(0) = 1$ is preserved exactly.

For a Voigt profile $g_{\text{src}}^{(1)}(\tau) = e^{-|\tau|/\tau_L} e^{-\tau^2/2\tau_G^2}$ — a Lorentzian of width τ_L (from optically-thick electron scattering or BLR turbulent broadening) convolved with a Gaussian of width τ_G (from bulk Doppler motions) — the jitter kernel combines with the source Gaussian into an effective width and mean,

$$\tilde{\sigma}_1^2 = \frac{\sigma_\Delta^2 \tau_G^2}{\sigma_\Delta^2 + \tau_G^2}, \quad \mu_1(\tau) = \frac{\tau_G^2 \tau}{\sigma_\Delta^2 + \tau_G^2}, \quad (14)$$

giving, for $\tau > 0$:

$$g_{\text{obs}}^{(1)}(\tau) = \frac{\tilde{\sigma}_1}{\sigma_\Delta} e^{-\tau^2/2(\sigma_\Delta^2 + \tau_G^2)} e^{\tilde{\sigma}_1^2/2\tau_L^2} \frac{1}{2} \left[e^{-\mu_1/\tau_L} \operatorname{erfc}\left(\frac{\tilde{\sigma}_1/\tau_L - \mu_1/\tilde{\sigma}_1}{\sqrt{2}}\right) + e^{\mu_1/\tau_L} \operatorname{erfc}\left(\frac{\tilde{\sigma}_1/\tau_L + \mu_1/\tilde{\sigma}_1}{\sqrt{2}}\right) \right]. \quad (15)$$

In the weak-jitter limit $\sigma_\Delta \ll \min(\tau_L, \tau_G)$, expanding Eq. (10) to second order in Δ and averaging over the Gaussian jitter gives the Voigt analogue of Eq. (13):

$$g_{\text{obs}}^{(1)}(\tau) \approx g_{\text{src}}^{(1)}(\tau) \left[1 + \frac{\sigma_\Delta^2(\tau)}{2} \left(\left(\frac{1}{\tau_L} + \frac{\tau}{\tau_G^2} \right)^2 - \frac{1}{\tau_G^2} \right) + \mathcal{O}(\sigma_\Delta^4) \right], \quad (16)$$

which reduces to Eq. (13) in the purely Lorentzian limit $\tau_G \rightarrow \infty$. Unlike the Lorentzian case, the correction factor can change sign: for $\tau < \tau_G \sqrt{1 - \tau_G/\tau_L}$ the second-order term is negative, so foam jitter *suppresses* short-lag coherence. Note that both closed forms in the weak-jitter limit share the general product structure $g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) F_{\text{foam}}(\tau)$.

C. Narrow-Band Limit

For a quasi-monochromatic source with center frequency ω_0 and angular-frequency bandwidth $\Delta\omega \ll \omega_0$, $E_{\text{src}}(t) = A(t) e^{-i\omega_0 t}$ where $A(t)$ varies on the coherence timescale $\tau_c = 1/\Delta\omega = \lambda^2/(2\pi c \Delta\lambda)$. When $\sigma_\ell/(c\tau_c) \ll 1$, typical realizations of δt leave the envelope unaffected, $A(t - \delta t) \approx A(t)$, and the jitter acts purely as a carrier phase modulation; the observed field takes the form

$$E_{\text{obs}}(t) \approx A(t) e^{-i\omega_0 t} e^{i\omega_0 \delta t(t)}. \quad (17)$$

(Corrections to the envelope approximation are of order $(\sigma_\ell/c\tau_c)^2$; for $\alpha = 1/2$ at $D = 1$ Gpc, $\lambda = 500$ nm, $\Delta\lambda/\lambda = 10^{-3}$ this parameter is ≈ 0.28 , so the exact convolution Eq. (12) should be used instead. For $\alpha = 2/3$ the analogous parameter is $\sim 10^{-11}$ (Section ??), so Eq. (19) is exact to any practical precision in the holographic case.) Since $A(t)$ and $\delta t(t)$ are independent, the first-order coherence factorizes. The jitter contribution is the characteristic function of $\Delta(\tau)$ evaluated at ω_0 ; since $\Delta(\tau)$ is Gaussian (Section II C),

$$\langle e^{i\omega_0 \Delta(\tau)} \rangle_{\delta t} = e^{-\omega_0^2 \sigma_\Delta^2(\tau)/2}. \quad (18)$$

This holds when $\tau \gg \tau_{\text{foam}}$ so that foam-cell contributions are independent; in the dynamical limit $\tau_{\text{foam}} \ll \tau_c$ the condition is satisfied at all observationally relevant lags. Substituting $\omega_0 = 2\pi c/\lambda$ and Eq. (??):

$$g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) \cdot \exp(-\sigma_\phi^2 [1 - C_\alpha(\tau; D)]), \quad (19)$$

where the rms accumulated phase follows from the Ng–Perlman path-length model (1) via the phase conversion $\sigma_\phi = (2\pi/\lambda_{\text{obs}}) \sigma_\ell(D_C)$, with D_C the line-of-sight comoving distance [4]:

$$\sigma_\phi^2 = \left(\frac{2\pi}{\lambda_{\text{obs}}} \right)^2 A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha}. \quad (20)$$

We write the foam suppression factor — the quantity appearing in the product structure of Section IIIB — as

$$\mathcal{F}_{\text{foam}}(\tau) = e^{-\sigma_\phi^2[1-C_\alpha(\tau;D)]}, \quad (21)$$

so $g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) \cdot \mathcal{F}_{\text{foam}}(\tau)$. At $\tau = 0$: $\mathcal{F}_{\text{foam}}(0) = 1$. At $\tau \rightarrow \infty$: $\mathcal{F}_{\text{foam}}(\infty) = e^{-\sigma_\phi^2[1-C_\alpha(\infty;D)]}$, which equals $e^{-\sigma_\phi^2}$ in the dynamical limit ($C_\alpha \rightarrow 0$) and approaches unity for quasi-static foam ($C_\alpha \approx 1$). The narrow-band condition $\sigma_\ell/(c\tau_c) \ll 1$ is compatible with the weak-jitter expansion of Eq. (13) — the former controls envelope displacement, the latter the carrier-phase suppression — and when neither applies the full convolution Eq. (10) must be evaluated.

IV. IMPLICATIONS FOR SPECTROSCOPY

A. The Wiener–Khinchin Bridge

The foam-modified observed coherence function — given in general by the jitter-convolution integral Eq. (10) with closed forms Eqs. (12)–(15) — carries direct spectroscopic content: for a stationary field the power spectral density $S(\nu)$ and $g^{(1)}(\tau)$ are Fourier-transform pairs, implying

$$S_{\text{obs}}(\nu) = \mathcal{F}\{g_{\text{obs}}^{(1)}(\tau)\}(\nu). \quad (22)$$

The theorem applies because astronomical emission lines have finite linewidth (so the Fourier transform is well defined) and the foam-induced jitter is a stationary process independent of the source, so the observed field is stationary when the integration time exceeds the foam coherence time ($T_{\text{int}} \gg \tau_{\text{foam}}$). In the further limit where the foam decorrelates on long time scales ($C_\alpha(\tau) \rightarrow 0$ as $|\tau| \rightarrow \infty$, the dynamical regime), the foam factor acquires a nonzero asymptote $\mathcal{F}_{\text{foam}}(\tau) \rightarrow e^{-\sigma_\phi^2}$, generating a distributional $\delta(\nu)$ contribution that encodes the fraction $e^{-\sigma_\phi^2}$ of source flux propagating phase-coherently.

When the lag-domain product structure $g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) \mathcal{F}_{\text{foam}}(\tau)$ holds—the weak-jitter and narrow-band regimes—the convolution theorem applied to Eq. (22) yields

$$S_{\text{obs}}(\nu) = S_{\text{src}}(\nu) * \mathcal{K}_{\text{foam}}(\nu). \quad (23)$$

i.e. the intrinsic spectrum is convolved by a kernel $\mathcal{K}_{\text{foam}}(\nu) \equiv \mathcal{F}\{\mathcal{F}_{\text{foam}}\}(\nu)$ that depends on both the foam (through σ_ϕ^2 , C_α , τ_{foam}) and the source line.

B. Connection to Fringe Visibility and Strehl Ratio

The first-order coherence function $g_{\text{obs}}^{(1)}(\tau)$ derived above describes *temporal* coherence along a single line of sight, with the foam suppression governed by the temporal decorrelation $C_\alpha(\tau; D)$ of Eq. (21). The fringe visibility and Strehl ratio are the *spatial* counterparts: they probe phase variations between two points in the telescope aperture plane separated by a baseline B , with the suppression governed by an aperture-plane coherence function $C_\alpha^{\text{sp}}(B; D)$.

The structural equivalence between the temporal and spatial cases is not merely formal: in both, the foam suppression arises as the characteristic function of a zero-mean Gaussian differential random variable, with C_α (respectively C_α^{sp}) setting how that variable’s variance grows with separation. Temporally, the relevant random variable is the differential jitter $\Delta(\tau) = \delta t(t + \tau) - \delta t(t)$ between two arrival times along a single ray, with variance $\sigma_\Delta^2(\tau) = (2\sigma_\ell^2/c^2)[1 - C_\alpha(\tau; D)]$ (Eq. (??)). Spatially, the analogous random variable is the differential phase $\Delta\phi(B) = \phi(\mathbf{x} + \mathbf{B}) - \phi(\mathbf{x})$ between two points on the aperture plane separated by baseline B , with variance

$$\sigma_{\Delta\phi}^2(B) = 2\sigma_\phi^2[1 - C_\alpha^{\text{sp}}(B; D)]. \quad (24)$$

Applying the Gaussian characteristic-function identity of Eq. (18) in each domain yields suppression factors of identical form—one expressed in lag, the other in baseline:

$$\mathcal{F}_{\text{foam}}^{\text{sp}}(B) \equiv \exp\{-\tfrac{1}{2}\sigma_{\Delta\phi}^2(B)\} = \exp\{-\sigma_\phi^2[1 - C_\alpha^{\text{sp}}(B; D)]\}, \quad (25)$$

mirroring $\mathcal{F}_{\text{foam}}(\tau)$ of Eq. (21) under $C_\alpha(\tau; D) \leftrightarrow C_\alpha^{\text{sp}}(B; D)$. The shared subscript α further reflects that the same cumulative growth law of Eq. (1) underlies both decorrelations. As with its temporal counterpart, $\mathcal{F}_{\text{foam}}^{\text{sp}}$ runs from unity at zero baseline ($C_\alpha^{\text{sp}}(0; D) = 1$, the two points sample identically phased wavefronts) to maximum suppression $e^{-\sigma_\phi^2}$ in the resolved limit where $C_\alpha^{\text{sp}} \rightarrow 0$ (the two points sample independent foam realizations).

The two-element interferometric fringe visibility at baseline B is then

$$V(B) = V_{\text{src}}(B) \mathcal{F}_{\text{foam}}^{\text{sp}}(B) = V_{\text{src}}(B) \exp\{-\sigma_\phi^2[1 - C_\alpha^{\text{sp}}(B; D)]\}, \quad (26)$$

where $V_{\text{src}}(B) = |g_{\text{src}}^{(1)}|$ is evaluated at angular scale λ/B . The Strehl ratio of a single aperture of diameter D_{tel} is the same statistic with B replaced by the internal aperture baseline: when $C_\alpha^{\text{sp}}(D_{\text{tel}}; D) \rightarrow 0$ the aperture-plane phases are independent draws from the single-ray distribution and the Maréchal formula gives [4]

$$S \simeq e^{-\sigma_\phi^2}. \quad (27)$$

Strehl ratio and fringe visibility therefore measure the *same* underlying phase variance $\sigma_\phi^2 = (2\pi\sigma_\ell/\lambda)^2$ in the resolved limit, differing only in whether what part of the point-spread-function is being considered. This unification places the published imaging constraints [4, 7–9] on a common footing within the present framework.

Equation (26) also extends the binary Lieu–Hillman [7] erasure criterion — their condition $\Delta\phi \gtrsim 2\pi$ (equivalently $\sigma_\phi \gtrsim 1$) marks the saturation limit $e^{-\sigma_\phi^2} \rightarrow 0$ of the fringe suppression — to a continuous curve that encodes the full aperture-plane coherence profile. The same saturated-limit expression $e^{-\sigma_\phi^2}$ is reinterpreted in Section VB as the coherent fraction of a spectral line, yielding a spectroscopic counterpart of the Strehl bound; the rapid chromatic tightening $\sigma_\phi \propto 1/\lambda$ that drives Ng & Perlman’s [4] Table 1 from optical to X-ray wavelengths is the same scaling exploited there.

V. SOURCES

Time-varying spacetime foam modifies the source first-order coherence, and hence its spectrum, in a wavelength-dependent way through the scaling $\sigma_\phi \propto 1/\lambda$. Two observable consequences follow. First, individual spectral features are blurred, in direct analogy to the blurring of the point-spread function in imaging. Second, the broadband spectral energy distribution is distorted: the cosmic microwave background, expected to be a near-perfect blackbody, would acquire foam-induced deviations whose chromatic signature is distinct from standard chemical-potential and Compton- y distortions. In this section we consider these two possibilities.

The two-component line-profile decomposition of Eq. (23) admits a direct application as a spectroscopic analogue of the Strehl-ratio bound [4]: just as the observation of a diffraction-limited point-spread function at wavelength λ and distance D requires the Strehl ratio $S \simeq e^{-\sigma_\phi^2}$ (Eq. (27)) to remain of order unity, i.e. $\sigma_\phi^2 \lesssim \mathcal{O}(1)$ (§3 and Eq. (4) of Ref. [4]), the observation of sharp spectral features requires the same condition: in the dynamical foam limit the lag-asymptote $\mathcal{F}_{\text{foam}}(\infty) = e^{-\sigma_\phi^2}$ of Eq. (21) plays the role of a *spectral Strehl* — the fraction of source flux delivered into the coherent line core $e^{-\sigma_\phi^2} S_{\text{src}}(\nu)$ of Eq. (23) — and for $\sigma_\phi^2 \gtrsim 1$ this coherent core is suppressed below the incoherent pedestal so that crisp line profiles cease to be detectable. The two arguments are structurally identical: each is a single-number, saturated-limit test of the coherent fraction $e^{-\sigma_\phi^2}$, differing only in whether that fraction is read off the on-axis PSF intensity or off the spectral line core. They share the same threshold criterion and the same parametric dependence on $\{D, \lambda, \alpha\}$.

A. GRB 221009A

GRB 221009A (“the BOAT,” Brightest Of All Time) is the most energetic gamma-ray burst in the instrumental record, with isotropic-equivalent energy $E_{\text{iso}} \sim 10^{55}$ erg at redshift $z = 0.151$, a once-in- $\sim 10^4$ -year event [13]. The burst was detected from radio wavelengths through ≈ 18 TeV (LHAASO). Two independent spectral analyses have reported a narrow gamma-ray emission feature in the Fermi/GBM data, interpreted as blueshifted electron–positron pair annihilation [14, 15], at observed energies in the range 10–37 MeV. We apply the spectral-Strehl bound Eq. (31) to each detection. Adopting flat Λ CDM ($\Omega_M = 0.27$, $\Omega_\Lambda = 0.73$, $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$), the comoving distance is $D_C(z = 0.151) \approx 0.63 \text{ Gpc} \approx 1.93 \times 10^{25} \text{ m}$ and $\ln(D_C/\ell_P) \approx 138.3$.

Ravasio et al. (2024). Ravasio et al. [14] report a 13σ detection of a narrow emission feature in Fermi/GBM data from the intervals $T + 280$ – 320 s, outside the bad-time interval affected by detector saturation. The line centroid evolves from $\approx 14 \text{ MeV}$ at $T + 280$ s to $\approx 6 \text{ MeV}$ at $T + 320$ s; the most energetic well-constrained single measurement

is $E_{\text{line}} = 12.56 \pm 0.30$ MeV in the interval $T + 280\text{--}300$ s (improvement $\Delta\text{AIC} = 49$ over the continuum-only model). The probe wavelength is $\lambda_{\text{obs}} = hc/E_{\text{line}} \approx 9.9 \times 10^{-14}$ m. The Gaussian width measured in this interval is $\sigma_{\text{gauss}} = 1.31^{+0.30}_{-0.31}$ MeV, giving a relative width $\sigma/E \approx 10\%$. The narrow-band condition underlying Eq. (23) is satisfied marginally: at the bound $\alpha \approx 0.64$ the rms path-length $\sigma_\ell \approx \lambda_{\text{obs}} \approx 9.9 \times 10^{-14}$ m by definition of the threshold, while the coherence length $c\tau_c \approx \hbar c/\sigma_E \approx 1.5 \times 10^{-13}$ m for $\sigma_{\text{gauss}} = 1.31$ MeV, giving $\sigma_\ell/(c\tau_c) \approx 0.66$; a precision bound would use the full convolution Eq. (12). Inserting $D_C \approx 1.93 \times 10^{25}$ m, $\lambda_{\text{obs}} \approx 9.9 \times 10^{-14}$ m, and $\ell_P = 1.6 \times 10^{-35}$ m into Eq. (31): $\ln(D_C/\lambda_{\text{obs}}) \approx 88.2$ and $\ln(D_C/\ell_P) \approx 138.3$, giving

$$\alpha \gtrsim \frac{88.2}{138.3} \approx 0.64. \quad (28)$$

Zhang et al. (2024). Zhang et al. [15] perform a joint analysis of GECAM-C and Fermi/GBM data covering the full prompt emission, including the bright phase inaccessible to GBM alone. They identify a spectral excess in the G1 time interval ($T + 246\text{--}256$ s) at ≈ 37 MeV, and interpret the entire feature as a line whose centroid evolves from ≈ 37 MeV down to ≈ 6 MeV across the burst. The 37 MeV detection lies at or beyond the stated upper boundary of the GBM BGO detector range (4–35 MeV), so the centroid and width of this earliest interval carry larger systematic uncertainty than the Ravasio et al. detections. Taking $E_{\text{line}} \approx 37$ MeV at face value gives $\lambda_{\text{obs}} \approx 3.4 \times 10^{-14}$ m, $\ln(D_C/\lambda_{\text{obs}}) \approx 89.2$, and

$$\alpha \gtrsim \frac{89.2}{138.3} \approx 0.65. \quad (29)$$

This bound should be regarded as tentative pending confirmation of the 37 MeV excess as a narrow spectral line within the instrumental sensitivity range.

Both bounds fall below but do not exclude the holographic value $\alpha = 2/3 \approx 0.667$: at that exponent $\sigma_\phi^2(\alpha = 2/3) \approx 0.01$ for the Ravasio line, corresponding to a coherent fraction $e^{-0.01} \approx 99\%$ — the line would be essentially unaffected. Formally ruling out $\alpha = 2/3$ at $z = 0.151$ under the Ng-Perlman criterion would require a confirmed spectral line at $E_{\text{obs}} \gtrsim 720$ MeV, approximately twenty times higher than the most energetic detection reported to date.

B. BzK4892

BzK 4892 is a massive ($M_\star \approx 5 \times 10^{10} M_\odot$), infrared-luminous, dust-obscured star-forming *BzK* galaxy in the CDF-S at spectroscopic redshift $z_s = 2.578$, hosting a heavily obscured, intrinsically quasar-luminous ($L_{2\text{--}10\text{ keV}} \gtrsim 10^{44.5} \text{ erg s}^{-1}$) AGN behind a Compton-thick absorber with $N_{\text{H}} \sim 10^{24.5\text{--}25} \text{ cm}^{-2}$ [16]. Because the direct hard X-ray continuum is suppressed by photoelectric absorption and Compton down-scattering, the X-ray spectrum is reflection-dominated and the 6.4 keV Iron K α fluorescence line—produced by the obscurer reprocessing the buried AGN continuum—emerges with exceptionally large rest-frame equivalent width, $\text{EW} \simeq 2.3$ keV, detected at $\gtrsim 4\sigma$ in the *Chandra* 4 Ms CDF-S exposure [16]. The redshift is independently confirmed by multiple narrow rest-UV emission lines (Ly α , C IV, He II, C III) detected in VLT/FORS spectroscopy [17, 18]. Adopting the same flat Λ CDM cosmology, the line-of-sight comoving distance is $D_C(z_s = 2.578) \approx 6.1$ Gpc $\approx 1.88 \times 10^{26}$ m. The spectroscopically detected Iron K α 6.4 keV emission line, redshifted to an observed energy of $6.4/(1+z_s) \approx 1.79$ keV ($\lambda_{\text{obs}} \approx 6.93 \times 10^{-10}$ m), provides the probe wavelength for this estimate.

The threshold condition $\sigma_\phi^2 \lesssim 1$ for the coherent line core of Eq. (23) to remain detectable above the foam-broadened pedestal — the spectral-Strehl criterion introduced above, mirroring the imaging Strehl bound of Eq. (27) — depends on the intrinsic source profile $S_{\text{src}}(\nu)$ only through its characteristic width $\Delta\nu_{\text{src}}$, and not through its detailed functional form. For the fluorescent Iron K α line in BzK 4892 the natural width and fine-structure splitting are negligible compared to the velocity broadening of the Compton-thick reprocessor ($\text{FWHM} \sim 10^3\text{--}10^4 \text{ km s}^{-1}$, i.e. $\Delta\nu_{\text{src}}/\nu_{\text{src}} \sim 10^{-3}\text{--}10^{-2}$), and the line is in any case instrumentally unresolved at *Chandra*/ACIS spectral resolution. We therefore adopt the narrow-line idealisation $\Delta\nu_{\text{src}} \ll \nu_{\text{src}}$, under which the coherent core and the foam-broadened pedestal of Eq. (23) are spectrally separable; profile refinements (Voigt wings, Compton shoulder) would shift the threshold $\sigma_\phi^2 \lesssim 1$ by at most factors of order unity.

In the dynamical foam limit the suppression factor Eq. (21) reduces to $\mathcal{F}_{\text{foam}}(\infty) = e^{-\sigma_\phi^2}$, which is the coherent fraction of the observed line profile via Eq. (23). Following Ng & Perlman [4], the threshold for detecting the Iron K α feature is that the foam path-length jitter remains smaller than the observed wavelength: $\sigma_\ell \lesssim \lambda_{\text{obs}}$, equivalently $\sigma_\phi \lesssim 2\pi$. Beyond this threshold the rms path-length perturbation spans more than one full oscillation and the coherent line core is effectively erased. The cumulative phase variance follows from the Ng-Perlman path-length model (1) via the global phase conversion $\sigma_\phi = 2\pi\sigma_\ell(D_C)/\lambda_{\text{obs}}$. The narrow-band condition $\sigma_\ell/(c\tau_c) \ll 1$ underlying Eq. (23) is

satisfied here: at $\alpha = 0.58$ one has $\sigma_\ell \approx \lambda_{\text{obs}} \approx 6.9 \times 10^{-10}$ m (the threshold value by definition) while the X-ray coherence length $c\tau_c = \lambda_{\text{obs}}^2 / (2\pi\Delta\lambda) \approx 3 \times 10^{-8}$ m for FWHM $\sim 10^3$ km s $^{-1}$, giving $\sigma_\ell / (c\tau_c) \approx 2 \times 10^{-2}$; the envelope approximation is accurate to better than 10^{-3} . The phase variance, reproducing Ng & Perlman [4] Eq. (4), is

$$\sigma_\phi^2 \simeq (2\pi)^2 \frac{D_C^{2(1-\alpha)} \ell_P^{2\alpha}}{\lambda_{\text{obs}}^2}, \quad (30)$$

i.e., Eq. (20) with $A_\alpha = 1$ as in Ref. [4], where D_C is the comoving distance per Ng & Perlman Eq. (5). Imposing the threshold $\sigma_\phi^2 \lesssim 1$ and inverting,

$$\alpha \gtrsim \frac{\ln(D_C/\lambda_{\text{obs}})}{\ln(D_C/\ell_P)}, \quad (31)$$

the spectroscopic analogue of Eq. (8) of Ref. [4]. Inserting the numerical values $D_C \approx 1.88 \times 10^{26}$ m, $\lambda_{\text{obs}} \approx 6.93 \times 10^{-10}$ m, and $\ell_P = 1.6 \times 10^{-35}$ m gives $\ln(D_C/\lambda_{\text{obs}}) \approx 81.6$ and $\ln(D_C/\ell_P) \approx 140.6$, hence $\alpha \gtrsim 0.58$.

This bound, $\alpha \gtrsim 0.58$, directly reproduces the Ng & Perlman [4] formulation: applying the same formula at optical wavelengths ($\lambda = 5000$ Å, $D = 10$ Gpc) gives $\alpha \gtrsim 0.53$, matching their Table 1 entry. The improvement from the optical to X-ray constraint is wavelength-driven: because $\sigma_\phi^2 \propto \lambda_{\text{obs}}^{-2}$, using the X-ray FeK α line at $\lambda_{\text{obs}} \approx 6.93 \times 10^{-10}$ m instead of a typical optical line at $\sim 5 \times 10^{-7}$ m increases the phase variance by a factor $\sim 5 \times 10^5$, making the threshold $\sigma_\ell \lesssim \lambda_{\text{obs}}$ far more constraining. The result does not rule out the holographic value $\alpha = 2/3$, consistent with all current optical and infrared imaging constraints. The bound $\alpha \gtrsim 0.58$ is weaker than the gamma-ray results of Section V A ($\alpha \gtrsim 0.64$ – 0.65) despite the much greater comoving distance ($D_C = 6.1$ Gpc vs. 0.63 Gpc). The gain for the gamma-ray probe is wavelength-driven: λ_{obs} for the 12.56 MeV line is ≈ 7000 times shorter than the Iron K α probe, boosting $\ln(D_C/\lambda_{\text{obs}})$ by ≈ 8.8 while the reduced distance decreases it by only ≈ 2.3 , for a net gain of ≈ 6.6 in the numerator of Eq. (31).

C. CMB

For a broadband thermal source such as the CMB, foam does not introduce a spectral-line shift but instead redistributes power: the coherent-core component is suppressed by $e^{-\sigma_\phi^2(\nu)}$ and the missing flux reappears in a broad pedestal whose shape depends on the short-time behaviour of the correlation function $C_\alpha(\tau)$. Because $\sigma_\phi^2 \propto \nu^2$ (Eq. (20)), the foam deficit is concentrated at high frequencies while the pedestal fills in at low frequencies, producing a spectral signature distinct from standard chemical-potential (μ) and Compton- y distortions.

The Fisher-matrix analysis treats four spectral parameters $\theta = (A_{\text{eff}}, T_0, \mu, y)$, where $A_{\text{eff}} = A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha}$ is the combined foam amplitude and $\sigma_\phi^2(\nu) = (2\pi\nu/c)^2 A_{\text{eff}}$. The linearized foam spectral template is

$$\frac{\partial I_\nu}{\partial A_{\text{eff}}} = -\left(\frac{2\pi\nu}{c}\right)^2 B_\nu(T_0) + K_\nu^{\text{fast}} \int_{\nu_{\text{min}}}^{\nu_{\text{max}}} \left(\frac{2\pi\nu'}{c}\right)^2 B_{\nu'}(T_0) d\nu', \quad (32)$$

comprising a high-frequency deficit proportional to $\nu^2 B_\nu$ and a compensating low-frequency pedestal whose shape is set by the decorrelation kernel K_ν^{fast} . A fiducial kernel $K_\nu^{\text{fast}} \propto (\nu \ln[\nu_{\text{max}}/\nu_{\text{min}}])^{-1}$ (uniform energy redistribution in $\ln \nu$) is adopted here. With this choice the pedestal integral $P_{\text{foam}} \equiv \int_{\nu_{\text{min}}}^{\nu_{\text{max}}} (2\pi\nu'/c)^2 B_{\nu'}(T_0) d\nu'$ is a frequency-independent constant, so the pedestal has a $1/\nu$ spectral shape — the same Rayleigh–Jeans dependence as the μ -distortion template, and hence the origin of the foam– μ degeneracy. The physical pedestal is determined by the short- τ behaviour of $C_\alpha(\tau)$ (Appendix A, Section A 3); the fiducial kernel is the simplest choice consistent with energy conservation. The standard CMB spectral distortions describe two physically distinct processes. The chemical-potential (μ) distortion arises from early-universe energy injection (e.g., Silk damping or decaying particles) when Compton scattering thermalizes photon energies but cannot adjust photon number, yielding a Bose–Einstein spectrum with $\mu \neq 0$. The Compton- y distortion encodes late-time electron heating (e.g., during reionization or in galaxy clusters) when full thermalization is no longer possible; it is proportional to the line-of-sight integral of electron thermal pressure. The linearized distortion templates are

$$\frac{\partial I_\nu}{\partial \mu} = -B_\nu(T_0) \frac{xe^x}{(e^x - 1)^2}, \quad (33)$$

$$\frac{\partial I_\nu}{\partial y} = B_\nu(T_0) \frac{xe^x}{e^x - 1} \left[\frac{x(e^x + 1)}{e^x - 1} - 4 \right], \quad (34)$$

where $x = h\nu/(k_B T_0)$. The foam- μ degeneracy arises because in the Rayleigh-Jeans limit ($x \ll 1$) the μ -template approaches $\Delta I/B \rightarrow -\mu/x \propto \nu^{-1}$, the same frequency dependence as the fiducial pedestal K_ν^{fast} ; the two templates are therefore nearly collinear across the FIRAS band. The temperature-bias entries in Fig. 2 follow from the nested-model bias formula $\Delta\theta_{\text{fit}} = -(F_{\text{fit,fit}})^{-1} F_{\text{fit,omit}} A_{\text{eff}}^{\text{true}}$, where $F_{\text{fit,fit}}$ is the 3×3 Fisher block for (T_0, μ, y) and $F_{\text{fit,omit}}$ is its cross-coupling to A_{eff} . The positive sign ($b_{T_0} = +1.21 \times 10^7 \text{ K m}^{-2}$) reflects the fact that the pedestal excess at low frequencies mimics a temperature increase in the Rayleigh-Jeans regime.

The CMB monopole spectrum measured by COBE/FIRAS over 60–600 GHz constrains spectral distortions at $|\mu|, |y| \lesssim 10^{-5} - 10^{-4}$ [19]. At the last-scattering surface ($D_C \approx 14 \text{ Gpc}$, Planck 2018 cosmology), the foam phase variance at 150 GHz is $\sigma_\phi^2 \approx 0.07$ for the random-walk model ($\alpha = 1/2$) and $\sigma_\phi^2 \approx 10^{-22}$ for the holographic model ($\alpha = 2/3$). A Fisher-matrix analysis of the FIRAS band (43 uniform channels, diagonal noise $\sigma_\nu = 0.01 \text{ MJy/sr}$) shows that the foam amplitude A_{eff} is nearly degenerate with the μ -distortion template (correlation coefficient $\rho = 0.976$, Eq. (35)), degrading the sensitivity by a factor of ~ 6 relative to the case where μ and T_0 are known. Accounting for this degeneracy, FIRAS is sensitive to foam only for $\alpha \lesssim 0.57$ (Fig. 1).

The Fisher-matrix correlation coefficients for $\theta = (A_{\text{eff}}, T_0, \mu, y)$ are

$$\rho_{ij} = \begin{pmatrix} 1.000 & 0.937 & \mathbf{0.976} & 0.010 \\ 0.937 & 1.000 & 0.969 & -0.212 \\ \mathbf{0.976} & 0.969 & 1.000 & -0.127 \\ 0.010 & -0.212 & -0.127 & 1.000 \end{pmatrix}, \quad (35)$$

showing the dominant foam- μ degeneracy ($\rho_{A_{\text{eff}}\mu} = 0.976$, bold) and negligible foam- y correlation ($\rho_{A_{\text{eff}}y} = 0.010$). The dominant degeneracy will shift once a physical C_α model replaces the fiducial pedestal kernel.

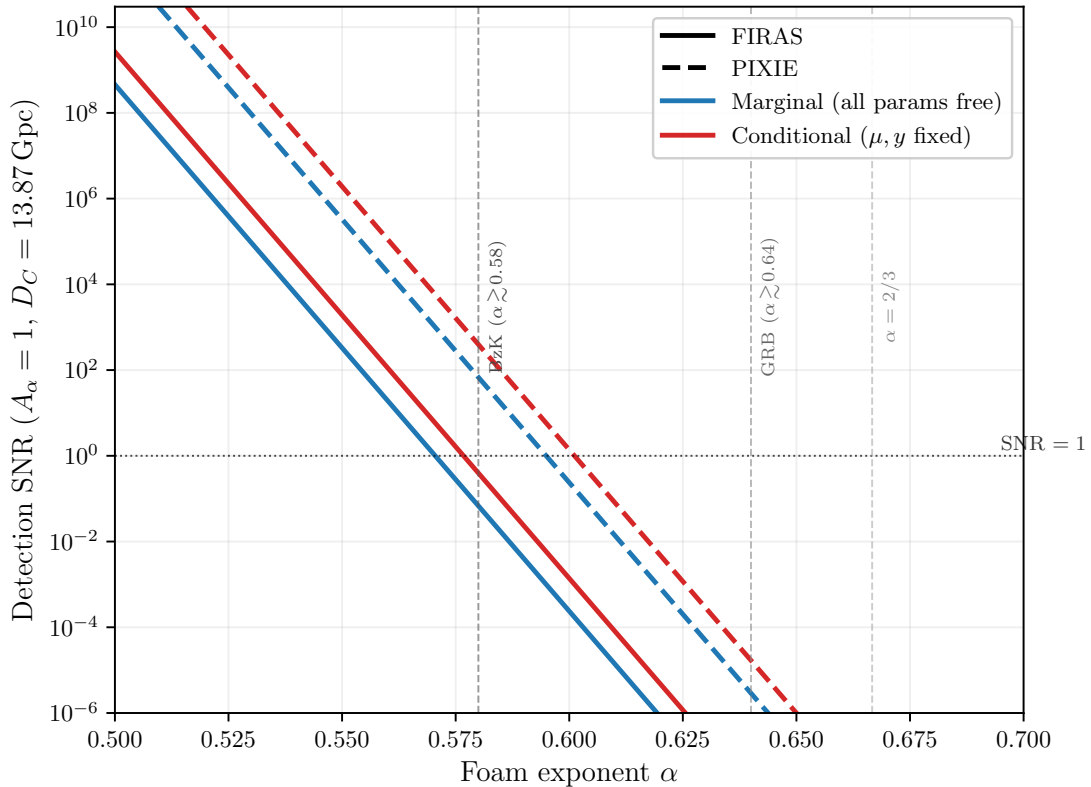


FIG. 1. Detection SNR for the foam amplitude A_{eff} as a function of foam exponent α ($A_\alpha = 1$, $D_C = 13.87 \text{ Gpc}$). Blue curves: marginal SNR (all four parameters $A_{\text{eff}}, T_0, \mu, y$ free). Red curves: conditional SNR (μ, y held fixed), which is $\sim 6\times$ higher. Solid: FIRAS; dashed: PIXIE (noise floor $\sim 10^3\times$ lower). The dotted horizontal line marks $\text{SNR} = 1$; vertical dashed lines mark the spectroscopic lower bounds from BzK 4892 ($\alpha \gtrsim 0.58$) and GRB 221009A ($\alpha \gtrsim 0.64$), and the holographic prediction $\alpha = 2/3$.

This sensitivity range ($\alpha \lesssim 0.57$) does not overlap with the spectroscopic lower bounds derived in Sections V A and V B: FIRAS probes strictly lower exponents than the two astrophysical sources studied above. Any foam model

consistent with the spectroscopic bounds ($\alpha \gtrsim 0.58$ – 0.65) produces a phase variance $\sigma_\phi^2 \ll 1$ at CMB frequencies; if such a model were present but omitted from the spectral fit, the recovered CMB temperature would be biased by $|\Delta T_0| \ll \sigma(T_0)$ (Fig. 2). Spacetime foam in this class therefore cannot account for any systematic error in the CMB temperature measurement or in cosmological quantities derived from it such as the Hubble constant H_0 .

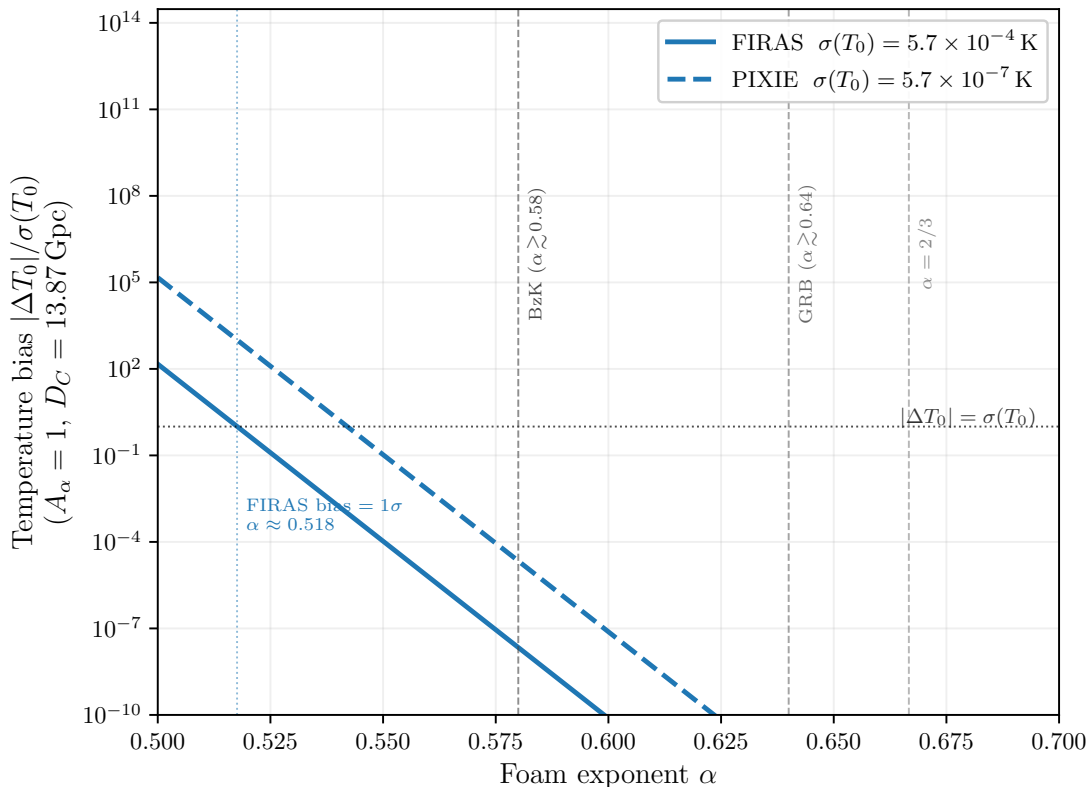


FIG. 2. Foam-induced bias in the recovered CMB temperature, $|\Delta T_0|/\sigma(T_0)$, as a function of foam exponent α when foam at amplitude A_{eff} ($A_\alpha = 1$, $D_C = 13.87$ Gpc) is present but omitted from the spectral fit. Solid: FIRAS with $\sigma(T_0) = 5.7 \times 10^{-4}$ K [20]; dashed: PIXIE with $\sigma(T_0) \approx 5.7 \times 10^{-7}$ K (noise floor $\sim 10^3 \times$ lower). The dotted horizontal line marks $|\Delta T_0| = \sigma(T_0)$; the FIRAS crossover occurs at $\alpha \approx 0.518$. Vertical dashed lines mark the spectroscopic lower bounds and the holographic prediction $\alpha = 2/3$ (as in Fig. 1).

A definitive FIRAS foam analysis would also require specifying $C_\alpha(\tau)$ to determine the pedestal shape, and replacing the linearized template with the full nonlinear expression at high frequencies where $\sigma_\phi^2 \gtrsim 1$ (Appendix A). FIRAS provides an independent check via absolute-calibrated broadband photometry — complementary to narrow spectral-line spectroscopy — and constrains the parameter space below both spectroscopic bounds where more exotic foam models would reside.

VI. CONCLUSIONS

A principal instrumental limitation of the present analysis is the MeV sensitivity gap: the spectral region between the upper edge of the Fermi-GBM BGO detector response ($\lesssim 40$ MeV, where the effective area falls steeply) and the regime in which Fermi-LAT achieves robust detection efficiency ($\gtrsim 100$ MeV). The frequency $\nu_\star \simeq 37$ MeV at which the foam-induced spectral break is predicted to be most constraining falls squarely within this gap, so neither instrument provides fully resolved coverage across the critical transition. Closing the MeV gap is a recognized community priority, addressed by next-generation satellite concepts that combine Compton-scattering and pair-conversion detection techniques to achieve continuous sensitivity from ~ 100 keV to beyond 1 GeV. The All-sky Medium Energy Gamma-ray Observatory (AMEGO [21]) and its MIDEX-scale variant AMEGO-X [22] are designed to cover 100 keV to 1 GeV with order-of-magnitude improvements over both COMPTEL and Fermi-LAT in the MeV decade; the ESA-proposed M-class concept e-ASTROGAM [23] targets a comparable bandpass. The near-term NASA SMEX mission COSI (scheduled launch ~ 2027) will extend broadband sensitivity to 0.2–5 MeV, providing a valuable pathfinder,

though its upper energy bound falls well short of the 37 MeV regime and the gap therefore remains open. Realization of any of the broader-bandpass concepts would, for the first time, permit fully resolved spectral measurements of GRB prompt emission across the 37 MeV transition without the instrumental discontinuity that limits the present work, removing a major systematic and enabling substantially tighter tests of spacetime-foam models.

The proposed PIXIE satellite [24] — a fully cryogenic ($T_{\text{sys}} \approx 2.7$ K) nulling polarizing Fourier transform spectrometer spanning 30 GHz to 6 THz — is projected to surpass FIRAS sensitivity by roughly three orders of magnitude, reaching $|\mu| \lesssim 10^{-8}$ and $|y| \lesssim 10^{-9}$ and entering the regime of guaranteed Λ CDM spectral signals from Silk damping and reionization. Propagating this gain through the Fisher-matrix scaling derived in Section V C, the marginal foam detection threshold shifts from $\alpha \lesssim 0.57$ (FIRAS) to approximately $\alpha \lesssim 0.60$ (PIXIE), bringing the CMB probe into marginal contact with the spectroscopic lower bound $\alpha \gtrsim 0.58$ from BzK 4892 (Fig. 1): whereas FIRAS achieves $\text{SNR} \approx 0.004$ at $\alpha = 0.58$, PIXIE would reach $\text{SNR} \approx 4$ at the same exponent. The dominant foam- μ degeneracy ($\rho \approx 0.98$) persists on the linearized template, but PIXIE's extension to 6 THz accesses the nonlinear regime $\sigma_\phi^2 \gg 1$ for $\alpha \lesssim 0.55$, where the foam template diverges markedly from the μ -distortion shape and the degeneracy may be substantially reduced. Complementary concepts including PRISM [25] and BISOU would push sensitivity further; a generation of absolute spectrophotometry missions thus opens a distinct window onto foam parameter space below the spectroscopic bounds established in Sections V A and V B.

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Appendix A: FIRAS Constraint on the Foam Exponent

1. Scope and Physical Motivation

This appendix does not present a completed FIRAS constraint on α ; rather, it identifies the physical inputs, derives the spectral template that any such analysis would fit, and outlines the analysis steps required to obtain a bound. The Wiener–Khinchin decomposition of Eq. (23) applies to any stationary field attenuated by the foam scattering kernel, not only to narrow emission lines. The Cosmic Microwave Background (CMB), observed across $\nu \in [6, 600]$ GHz by COBE/FIRAS, provides an independent broadband laboratory for this test: the same two-component signature — a suppressed coherent core and an incoherent pedestal (Section IV) — should appear in the CMB monopole spectrum if spacetime foam distorts photons on their path from the last-scattering surface. Because the foam signal is achromatic (geometric path-length jitter, not plasma dispersion), the frequency dependence of the foam-induced distortion follows $\sigma_\phi^2 \propto \nu^2$, distinct from standard chemical-potential (μ) and Compton- y spectral distortions. FIRAS constrains such distortions at the level $|\mu|, |y| \lesssim 10^{-5}$ to 10^{-4} .

2. Foam Phase Variance at CMB Frequencies

CMB photons travel from the last-scattering surface at comoving distance $D_C \approx 14$ Gpc (Planck 2018 flat Λ CDM cosmology; $H_0 = 67.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$). Equation (20) gives the accumulated phase variance directly at the comoving distance $D_C \approx 14$ Gpc:

$$\sigma_\phi^2(\nu) = \left(\frac{2\pi\nu}{c} \right)^2 A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha}, \quad (\text{A1})$$

where ν is the observed frequency. The strong ν^2 dependence means foam effects are largest at the high-frequency end of the FIRAS band. Representative values at $\nu = 150$ GHz ($\lambda_{\text{CMB}} \approx 2$ mm):

$$\sigma_\phi^2(150 \text{ GHz}) \approx \begin{cases} 6.8 \times 10^{-2} & \alpha = 1/2 \text{ (random walk)}, \\ 10^{-22} & \alpha = 2/3 \text{ (holographic)}. \end{cases} \quad (\text{A2})$$

The strong ν^2 dependence of σ_ϕ^2 means foam effects are most pronounced at high frequency; for $\alpha = 1/2$ the variance reaches unity ($\sigma_\phi^2 = 1$) at $\nu \approx 574$ GHz, near the top of the FIRAS band, marking where the nonlinear regime becomes important. The achievable sensitivity threshold (accounting for degeneracies with μ and T_0) is $\alpha \lesssim 0.57$.

3. Derivation of the Observed CMB Spectrum

The narrow-band reduction of Eq. (23) — which underlies the spectral analysis of Section IV — was derived for quasi-monochromatic sources and does not apply to the broadband blackbody. Here we derive the foam signature for the CMB by starting from the full convolution integral and evaluating it in the fully dynamical limit $C_\alpha \rightarrow 0$.

Step 1: Saturated jitter variance

In the $C_\alpha \rightarrow 0$ limit, Eq. (??) gives

$$\sigma_\Delta^2(\tau) \rightarrow \frac{2\sigma_\ell^2(D)}{c^2} \quad \text{for all } \tau \neq 0. \quad (\text{A3})$$

The jitter variance saturates to a τ -independent value, twice the single-ray variance σ_ℓ^2/c^2 .

Step 2: Monochromatic decomposition

The blackbody field is broadband; we decompose it as an incoherent superposition of narrowband modes: $S_{\text{src}}(\nu') = B_{\nu'}(T_0)$ (spectral intensity per unit frequency). Each mode at frequency ν' is an independent stationary Gaussian process with coherence function

$$g_{\nu'}^{(1)}(\tau) = e^{-i2\pi\nu'\tau} \times [\text{envelope}(|\tau|)]. \quad (\text{A4})$$

Critically, each mode is independently phase-modulated by the *same* achromatic jitter $\delta t(t)$, which is frequency-independent by assumption (geometric path-length fluctuation, not plasma dispersion).

Step 3: Per-mode foam suppression

For a monochromatic mode at frequency ν' , the foam suppression factor is

$$\mathcal{F}_{\text{foam}}(\tau; \nu') = e^{-\sigma_\phi^2(\nu')[1-C_\alpha(\tau)]}, \quad \text{where } \sigma_\phi^2(\nu') = (2\pi\nu'/c)^2 \sigma_\ell^2(D). \quad (\text{A5})$$

In the $C_\alpha \rightarrow 0$ limit, $\mathcal{F}_{\text{foam}}(\tau; \nu') \rightarrow e^{-\sigma_\phi^2(\nu')}$ for all $\tau \neq 0$ — a constant, τ -independent suppression.

Step 4: Two-component decomposition

Decompose the suppression factor into its asymptotic and residual parts:

$$\mathcal{F}_{\text{foam}}(\tau; \nu') = e^{-\sigma_\phi^2(\nu')} + \left[\mathcal{F}_{\text{foam}}(\tau; \nu') - e^{-\sigma_\phi^2(\nu')} \right]. \quad (\text{A6})$$

The first term is the $\tau \rightarrow \infty$ asymptote (the coherent core fraction); the second (the residual) vanishes at large $|\tau|$.

Step 5: Total first-order coherence

The intensity-weighted integral over modes:

$$g_{\text{obs}}^{(1)}(\tau) = \frac{1}{I_{\text{tot}}} \int_0^\infty B_{\nu'}(T_0) e^{-i2\pi\nu'\tau} \mathcal{F}_{\text{foam}}(\tau; \nu') d\nu'. \quad (\text{A7})$$

Step 6: Wiener-Khinchin to the observed spectrum

Taking the Fourier transform via Eq. (22) and applying the two-component decomposition:

$$S_{\text{obs}}(\nu) = e^{-\sigma_\phi^2(\nu)} B_\nu(T_0) + \int_0^\infty B_{\nu'}(T_0) \hat{R}(\nu - \nu'; \nu') d\nu', \quad (\text{A8})$$

where $\hat{R}(\nu - \nu'; \nu')$ is the Fourier transform of the residual $\mathcal{F}_{\text{foam}}(\tau; \nu') - e^{-\sigma_\phi^2(\nu')}$.

Key observations:

- (i) The coherent core term $e^{-\sigma_\phi^2(\nu)} B_\nu(T_0)$ is exact and model-independent; it requires only the assumptions of achromatic jitter and $C_\alpha \rightarrow 0$.
- (ii) The pedestal kernel $\hat{R}(\nu - \nu'; \nu')$ depends on the *short- τ behavior* of $C_\alpha(\tau)$ — that is, on the foam coherence timescale τ_{foam} . Without specifying $C_\alpha(\tau)$, the pedestal shape is *undetermined*.
- (iii) The current placeholder kernel $K_\nu^{\text{fast}} = [\nu \ln(\nu_{\text{max}}/\nu_{\text{min}})]^{-1}$ does not correspond to any specific C_α model and should be replaced once one is chosen. The FIRAS bandpass [6, 600] GHz enters the analysis only as a likelihood window, not as the physical support of the redistribution kernel.

Photon-number conservation

The achromatic phase modulation $e^{i\omega\delta t(t)}$ at each frequency is unitary and time-reversal-invariant, so it conserves the photon number per mode: each photon is merely time-shifted, not absorbed or created. Therefore

$$\int_0^\infty \frac{S_{\text{obs}}(\nu) - B_\nu(T_0)}{h\nu} d\nu = 0 \quad (\text{photon-number conservation by achromatic unitarity}). \quad (\text{A9})$$

The two-component split $(e^{-\sigma_\phi^2} B_\nu) + \text{pedestal}$ satisfies this automatically if the pedestal is computed from the residual via Fourier transform.

Connection to Section IV.A

The derivation above is the broadband analogue of the assertion in Section IV.A that the complex amplitude decomposes as a $\delta(\nu)$ coherent-core term plus a broadened pedestal. A future revision should either lift this derivation into the main text or reference this appendix from Section IV.A for the broadband case.

4. Validity of the Fast-Foam Assumption

The fast-foam limit requires $\tau_{\text{foam}} \ll 1/\nu_{\text{CMB}} \sim 7$ ps. If the foam coherence time is set by the Planck scale, $\tau_{\text{foam}} \sim \ell_P/c \approx 5 \times 10^{-44}$ s, this condition is satisfied by forty orders of magnitude. For larger τ_{foam} , the foam suppression factor $e^{-\sigma_\phi^2[1-C_\alpha(\tau_{\text{foam}})]}$ (Eq. (21)) is reduced by the factor $[1 - C_\alpha(\tau_{\text{foam}})]$, and the distortion amplitude is suppressed accordingly. In the quasi-static limit $\tau_{\text{foam}} \rightarrow \infty$, $C_\alpha \rightarrow 1$ and the CMB spectrum is unchanged, consistent with the vanishing of the HBT Siegert signal in that limit (Section III). In this limit the pedestal (residual term) also vanishes, consistent with Section A 3.

5. Degeneracy with Standard Spectral Distortions

The foam signal must be separated from two standard CMB spectral distortion templates. The chemical-potential (μ) and Compton- y distortions have characteristic shapes:

$$\Delta I_\nu^\mu = -\mu B_\nu(T_0) \frac{xe^x}{(e^x - 1)^2}, \quad (\text{A10})$$

$$\Delta I_\nu^y = y B_\nu(T_0) \frac{xe^x}{e^x - 1} \left[x \frac{e^x + 1}{e^x - 1} - 4 \right], \quad (\text{A11})$$

with $x = h\nu/(k_B T_0)$ and $T_0 = 2.7255$ K.

Comparing to the linearized foam template (high- ν deficit $\propto \nu^2 B_\nu$, low- ν pedestal undetermined), a key degeneracy emerges: the μ -distortion template's Rayleigh–Jeans limit is $\Delta I/B \propto -\mu/x \propto \nu^{-1}$, which is shape-degenerate with a $1/\nu$ -distributed pedestal. The current placeholder kernel $K_\nu^{\text{fast}} \propto 1/(\nu \ln[\nu_{\text{max}}/\nu_{\text{min}}])$ is precisely this form, maximizing the foam- μ correlation.

A Fisher matrix analysis (code: `fisher_firas.py`) using the linearized templates, diagonal FIRAS noise $\sigma_\nu = 0.01$ MJy/sr per channel, and 43 uniform channels from 60 to 600 GHz gives the correlation matrix shown in Eq. (35) and the marginal uncertainties

$$\sigma(A_{\text{eff}}) = 1.5 \times 10^{-17} \text{ m}^2 \quad (\text{marginal, all params free}), \quad \sigma(A_{\text{eff}})|_{\mu, y \text{ fixed}} = 2.6 \times 10^{-18} \text{ m}^2. \quad (\text{A12})$$

Marginalizing over μ and T_0 therefore degrades the foam amplitude uncertainty by a factor of ~ 6 , driven almost entirely by the foam- μ degeneracy.

Eq. (35) and Fig. 1 (Section VC) give the full correlation matrix and the SNR as a function of α , showing that sensitivity falls below 1σ at $\alpha \approx 0.57$.

These results use the placeholder kernel; the actual sensitivity limit depends on the physical pedestal shape. The foam- y correlation $\rho = 0.010$ is negligible: the y -distortion measurement provides essentially no help in separating the foam signal.

Action item: Repeat this Fisher analysis using (i) the FIRAS noise-plus-calibration covariance matrix from Fixsen (1996) in place of the diagonal approximation, and (ii) the full nonlinear foam template once $C_\alpha(\tau)$ is specified. Both changes will alter the correlations quantitatively, particularly the foam- μ degeneracy.

6. Bias in Recovered T_0 from Unmodeled Foam

The Fisher matrix provides two complementary measures of sensitivity. Section A5 asked *how precisely can we detect A_{eff} ?*; here we ask *how much would the fitted CMB temperature shift if foam were present but omitted from the model?*

Bias formula. Suppose the data contain a foam signal at amplitude $A_{\text{eff}}^{\text{true}}$ but the analyst fits only the standard parameters $\theta_{\text{fit}} = (T_0, \mu, y)$. In the linearized Fisher regime the resulting parameter shift is

$$\hat{\theta}_{\text{fit}} - \theta_{\text{fit}}^{\text{true}} = -(F_{\text{fit,fit}})^{-1} F_{\text{fit,omit}} A_{\text{eff}}^{\text{true}}, \quad (\text{A13})$$

where $F_{\text{fit,fit}}$ is the 3×3 lower-right block of the full Fisher matrix (for T_0, μ, y) and $F_{\text{fit,omit}}$ is the 3×1 cross-block coupling the fitted parameters to the omitted foam amplitude. Both blocks are extracted from the same matrix as Eq. (35) and Fig. 1, so no new computation is required.

Temperature bias. The T_0 component of Eq. (A13) is

$$\Delta T_0 \equiv \hat{T}_0 - T_0^{\text{true}} = b_{T_0} A_{\text{eff}}^{\text{true}}, \quad b_{T_0} \equiv -[(F_{\text{fit,fit}})^{-1} F_{\text{fit,omit}}]_{T_0}. \quad (\text{A14})$$

Numerically, $b_{T_0} = +1.21 \times 10^7 \text{ K m}^{-2}$: the bias is *positive*, meaning foam would cause the fitter to report a temperature slightly too high. This is counterintuitive but physically consistent. The placeholder pedestal $K_\nu^{\text{fast}} \propto \nu^{-1}$ produces a low-frequency flux excess that the fitter interprets as a temperature increase (since $\partial B_\nu / \partial T > 0$ in the Rayleigh–Jeans regime), while the high-frequency deficit is absorbed into a positive $\Delta\mu$. The dominant bias is in μ : numerically $b_\mu \approx 1.05 \times 10^8 \text{ m}^{-2}$, consistent with the strong foam- μ correlation $\rho = 0.976$ from Eq. (35).

Bias-based bound. Requiring $|\Delta T_0| < \sigma(T_0) = 0.00057 \text{ K}$ (Fixsen 2009 [20]) gives

$$A_{\text{eff}}^{\text{max, bias}} = \frac{\sigma(T_0)}{|b_{T_0}|} = \frac{5.7 \times 10^{-4} \text{ K}}{1.21 \times 10^7 \text{ K m}^{-2}} = 4.7 \times 10^{-11} \text{ m}^2, \quad (\text{A15})$$

corresponding, with $A_\alpha = 1$, to a crossover exponent $\alpha_{\text{bias}} = 0.518$. Foam models with $\alpha < 0.518$ would have biased the Fixsen T_0 by more than 1σ had such a signal been present and unmodeled; models with $\alpha > 0.518$ do not.

Fig. 2 (Section VC) plots $|\Delta T_0|/\sigma(T_0)$ and its ratio to $\sigma(T_0)$ for $\alpha = 0.50$ – 0.59 . At $\alpha = 0.57$ (where the marginal SNR ≈ 1), the ratio $\Delta T_0/\sigma(T_0) \approx 4 \times 10^{-4}$: completely negligible. The published Fixsen $T_0 = 2.72548 \text{ K}$ is therefore not appreciably biased by any foam model near or above the FIRAS detection threshold.

Comparison of bounds. The three α lower bounds from FIRAS are: (i) $\alpha \gtrsim 0.52$ (T_0 -bias bound, this subsection); (ii) $\alpha \gtrsim 0.57$ (marginal-SNR bound, Section A5); (iii) $\alpha \gtrsim 0.58$ (BzK 4892 spectroscopic bound, Section VB). The bias bound is the weakest because the strong foam- μ correlation ($\rho = 0.976$) allows most of the foam signal to be

absorbed into μ without producing a large *individual* shift in T_0 . The same freedom that hides the bias inflates the marginal $\sigma(A_{\text{eff}})$, hence the SNR bound is tighter.

Caveat: The sign and magnitude of ΔT_0 depend on the placeholder pedestal kernel K_ν^{fast} . A physically derived pedestal from a specific $C_\alpha(\tau)$ model (Section A 3) may alter these values; repeating the bias calculation with the physical kernel is action item 2b of Section A 10.

7. Parameter Degeneracy Within the Foam Template

The foam template parameters α and A_α enter the linearized distortion only through the product $A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha}$ — defining the combined amplitude A_{eff} . FIRAS constrains this amplitude directly, but not the exponent α separately. Converting a bound on A_{eff} to a bound on α requires fixing A_α by convention (e.g., $A_\alpha = 1$ following Ng and Perlman). Any reported foam constraint on α must state this assumption explicitly.

8. Nonlinear Regime at the Band Edge

The linearization $\Delta I_\nu \approx \sigma_\phi^2(\nu) \times [\text{template}]$ is valid only when $\sigma_\phi^2 \ll 1$. For $\alpha = 1/2$ (random-walk foam), $\sigma_\phi^2 = 1$ is reached at $\nu \approx 574$ GHz, within the FIRAS band. At higher frequencies the linearization breaks down and the full nonlinear expression from Section A 3 must be used in the likelihood. The nonlinear curvature at high ν may help break the degeneracy with μ and y that dominates the marginal parameter space; this should be checked after computing the nonlinear Fisher matrix.

9. Relation to the BzK 4892 Bound

Section V B reports $\alpha \gtrsim 0.58$ from the Fe K α spectral-Strehl test at $z = 2.578$, and Section V A reports $\alpha \gtrsim 0.64$ – 0.65 from MeV gamma-ray emission lines at $z = 0.151$. The FIRAS constraint achieves marginal sensitivity at $\alpha \lesssim 0.57$. Neither spectroscopic range overlaps with the FIRAS-sensitive region; FIRAS probes strictly lower α values (longer propagation distance, higher cumulative variance at lower exponents). FIRAS therefore provides an independent systematics check using completely different inputs (absolute calibrated photometry, broadband rather than single-line, no source profile assumption) and probes the parameter space below both spectroscopic bounds where more exotic foam models would reside.

10. Proposed Analysis

The FIRAS constraint requires the following steps:

1. *Specify $C_\alpha(\tau)$ model.* Choose or derive the temporal decorrelation function (e.g., exponential with timescale τ_{foam} , power-law, or data-driven). Use Section A 3 to compute the pedestal kernel $\hat{R}$ from first principles.
2. *Compute nonlinear Fisher matrix.* Using the full nonlinear template (not the linearized approximation), compute the Fisher correlation matrix for $(\alpha, A_\alpha, T_0, \mu, y, A_{\text{CIB}}, c_i)$ on the FIRAS noise-plus-calibration covariance. Assess how much the α - μ degeneracy is reduced once C_α is specified.
- 2b. *Repeat bias calculation in the nonlinear regime.* Once the full nonlinear template is available, repeat the linearized bias computation of Section A 6 at the nonlinear level to verify that the sign and magnitude of ΔT_0 are reliable at the band edge ($\nu \gtrsim 400$ GHz, where $\sigma_\phi^2 \rightarrow 1$ for $\alpha \approx 0.50$ – 0.55). In the nonlinear regime the linearized formula Eq. (A13) must be replaced by a numerical comparison of maximum-likelihood estimators.
3. *Null-hypothesis validation.* Fit the FIRAS spectrum with $A_\alpha = 0$ and verify recovery of published values: $T_0 = 2.72548 \pm 0.00057$ K, $|\mu| < 9 \times 10^{-5}$, $|y| < 1.5 \times 10^{-5}$ (95% CL; Fixsen 1996, 2009).
4. *Joint Bayesian inference.* Sample the full posterior $p(\alpha, A_\alpha, T_0, \mu, y, \dots | \mathbf{I}^{\text{obs}})$ using the full nonlinear likelihood and FIRAS data. Report marginal bounds on α (fixing $A_\alpha = 1$ or marginalizing over it as specified).

11. Connection to the Spectral Framework of Section IV

Equation (??) in its original form asserts the two-component decomposition (coherent core plus pedestal) without derivation. The detailed calculation of Section A 3 is what grounds this decomposition: it shows how the broadband

blackbody, when phase-modulated by achromatic jitter in the dynamical limit $C_\alpha \rightarrow 0$, naturally splits into a coherent-core term (exact, model-independent) and a pedestal (model-dependent on C_α). The FIRAS measurement probes this structure using the full 6–600 GHz dynamic range, complementing the single-line spectroscopy of Sections V A–V B.

12. Likelihood and Inference Framework

The FIRAS monopole spectrum is compared to the full foam template (Section A 3) together with standard spectral-distortion templates (μ -distortion, Compton- y , CIB, and calibration modes) via a Gaussian log-likelihood:

$$\ln \mathcal{L} = -\frac{1}{2} \mathbf{r}^T \mathbf{C}^{-1} \mathbf{r}, \quad (\text{A16})$$

where $\mathbf{r} = \mathbf{I}^{\text{obs}} - \mathbf{I}^{\text{model}}$ is the residual vector and $\mathbf{C}$ is the FIRAS noise-plus-calibration covariance matrix (Fixsen 1996). The free parameters are $(\alpha, A_\alpha, T_0, \mu, y, A_{\text{CIB}}, c_1, c_2, c_3)$, where c_i are calibration mode amplitudes. A null-hypothesis validation fit with $A_\alpha = 0$ must recover the published FIRAS values $T_0 = 2.72548 \pm 0.00057 \text{ K}$, $|\mu| < 9 \times 10^{-5}$, $|y| < 1.5 \times 10^{-5}$ (95% CL) before the foam parameter is freed for analysis.

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- [1] J. A. Wheeler, *Physical Review* **97**, 511 (1955).
 - [2] S. W. Hawking, *Nuclear Physics B* **144**, 349 (1978).
 - [3] S. Carlip, *Rep. Prog. Phys.* **86**, 066001 (2023), arXiv:2209.14282 [gr-qc].
 - [4] Y. J. Ng and E. S. Perlman, *Universe* **8**, 382 (2022).
 - [5] Y. J. Ng, *Physical Review Letters* **86**, 2946 (2001).
 - [6] Y. J. Ng, *Modern Physics Letters A* **18**, 1073 (2003).
 - [7] R. Lieu and L. W. Hillman, *The Astrophysical Journal* **585**, L77 (2003).
 - [8] W. A. Christiansen, Y. J. Ng, D. J. E. Floyd, and E. S. Perlman, *Physical Review D* **83**, 084003 (2011).
 - [9] E. S. Perlman, S. A. Rappaport, W. A. Christiansen, Y. J. Ng, J. DeVore, and D. Pooley, *The Astrophysical Journal* **805**, 10 (2015).
 - [10] V. S. H. Lee, K. M. Zurek, and Y. Chen, *Physical Review D* **109**, 084005 (2024), arXiv:2312.06757 [gr-qc].
 - [11] E. Steinbring, *Galaxies* **11**, 115 (2023).
 - [12] E. Steinbring and Y. J. Ng, arXiv e-prints 10.48550/arXiv.2603.18277 (2026), arXiv:2603.18277.
 - [13] E. Burns *et al.*, *Astrophys. J. Lett.* **946**, L31 (2023), arXiv:2303.14172.
 - [14] M. E. Ravasio *et al.*, *Science* **385**, 452 (2024).
 - [15] Y.-Q. Zhang *et al.*, *Sci. China Phys. Mech. Astron.* (2024), arXiv:2403.12851.
 - [16] C. Feruglio, E. Daddi, F. Fiore, D. M. Alexander, E. Piconcelli, and C. Malacaria, *Astrophysical Journal Letters* **729**, L4 (2011), arXiv:1101.3478.
 - [17] G. P. Szokoly, J. Bergeron, G. Hasinger, I. Lehmann, L. Kewley, V. Mainieri, M. Nonino, P. Rosati, R. Giacconi, R. Gilli, R. Gilmozzi, C. Norman, M. Romaniello, E. Schreier, P. Tozzi, J. X. Wang, W. Zheng, and A. Zirm, *Astrophysical Journal Supplement Series* **155**, 271 (2004), arXiv:astro-ph/0312141.
 - [18] E. Vanzella, S. Cristiani, M. Dickinson, M. Giavalisco, H. Kuntschner, J. Haase, M. Nonino, P. Rosati, C. Cesarsky, H. C. Ferguson, R. A. E. Fosbury, A. Grazian, L. A. Moustakas, A. Rettura, P. Popesso, A. Renzini, D. Stern, and GOODS Team, *Astronomy & Astrophysics* **454**, 423 (2006), arXiv:astro-ph/0509665.
 - [19] D. J. Fixsen, E. S. Cheng, J. M. Gales, J. C. Mather, R. A. Shafer, and E. L. Wright, *Astrophysical Journal* **473**, 576 (1996), arXiv:astro-ph/9605054.
 - [20] D. J. Fixsen, *Astrophysical Journal* **707**, 916 (2009), arXiv:0911.1955.
 - [21] J. McEnery, A. van der Horst, A. Dominguez, A. Moiseev, A. Marcowith, A. Harding, A. Lien, A. Giuliani, A. Inglis, S. Ansoldi, *et al.*, in *Bulletin of the AAS*, Vol. 51 (2019) p. 245, astro2020 Decadal Survey APC White Paper, arXiv:1907.07558 [astro-ph.IM].
 - [22] R. Caputo, M. Ajello, C. A. Kierans, J. S. Perkins, J. L. Racusin, L. Baldini, E. Burns, P. Cristofari, P. Coppin, P. de la Torre Luque, R. Engel, H. Fleischhack, C. L. Fryer, F. Gargano, A. Giuliani, D. Hartmann, M. Kachelriess, C. Karwin, J. F. Krizmanic, T. R. Lewis, F. Loparco, A. Manfreda, M. N. Mazziotta, J. E. McEnery, S. Mereghetti, J. W. Mitchell, A. A. Moiseev, M. Negro, N. Omodei, E. Orlando, M. Persic, R. Pillera, T. A. Porter, A. Pozanenko, G. Puehlhofer, M. Razzano, C. Sgrò, P. Shawhan, C. Shrader, D. A. Smith, H. Tajima, V. Tatischeff, D. J. Thompson, J. Valverde, T. M. Venters, M. Wood, and G. Zaharijas, *Journal of Astronomical Telescopes, Instruments, and Systems* **8**, 044003 (2022), arXiv:2208.04990 [astro-ph.IM].
 - [23] A. De Angelis, V. Tatischeff, M. Tavani, U. Oberlack, I. Grenier, L. Hanlon, R. Walter, A. Argan, P. von Ballmoos, A. Bulgarelli, I. Donnarumma, M. Hernanz, I. Kuvvetli, M. Pearce, A. Zdziarski, J. Conrad, G. Ghisellini, A. Harding, J. Isern, M. Leising, F. Longo, G. Madejski, M. Martinez, M. N. Mazziotta, J. M. Paredes, M. Pohl, R. Rando, M. Razzano, A. Aboudan, M. Ackermann, *et al.*, *Experimental Astronomy* **44**, 25 (2017), arXiv:1611.02232 [astro-ph.HE].
 - [24] A. Kogut *et al.*, *JCAP* **2011** (07), 025, arXiv:1105.2044.
 - [25] P. André *et al.*, *JCAP* **2014** (02), 006, arXiv:1310.1554.